
Topology of Justice: Law, Rights, and the Geometry of Social Constraint

[T]-Theory Volume: Law and Jurisprudence

Alistair Johnson

Independent Researcher, Zurich, Switzerland

ORCID: [0009-0007-2194-0850](#)

Contents

Introduction: What Law Is Doing When It Works	3
Rights as Topological Invariants	4
The Rule of Law as Ergodicity	4
Legal Uncertainty as Attractor Fragmentation	5
Regulation as a Control-Theory Problem	5
What This Book Offers the Legal Scholar	6
The Programme	7
The Gap the Programme Addresses	7
The Structure of the Argument	8
The Method: Mathematical Co-identification	9
What It Is	9
Why It Matters as Method	9
When MCI Is Over: The Verification Threshold	10
The Model: The Soma-Field	12
Five Co-identifications	12
What the Model Predicts	13
The Empirical Test: QUANT-EXP-1	14
The Prediction	14
The Experiment	14
Results	14
The Penrose Connection	15
The Lived Case: Field Notes from the Inside	15
Extensions: Music, Film, and the Domain Generality of the Model	16
Music-Induced Affect	16
The Tensor: An Abstract Film	17
The Argument as a Whole	18
What Remains	19
Data and Code Availability	20
Introduction: The Missing Unification	21
The Green's Function as the Universal SHO	21
The String Theory Problem	22
The Identification	22
Consequences	23
The 11-Dimensional Architecture	23
The Decomposition	23
The Limbic Axis as the Horava-Witten Orbifold	24
The 20-Scale Dial	24
The Organism Hierarchy	25
Three Tiers	25

Consciousness as Phase Transition	26
The Classical Gap	26
The Threshold	26
What Consciousness Is	26
Relation to Existing Frameworks	27
McFadden’s CEMI Theory	27
Schreiber’s Modal Homotopy Type Theory	27
Hoffman’s Conscious Agents	27
Formal Verification	28
The Volitional Agent	29
From Autonomous to Driven Dynamics	29
The Somatic Injection	29
Patient to Pilot	29
Discussion	30
What Has Been Claimed	30
The Correspondence Principle at Every Scale	30
Lean 4 as Epistemological Standard	31
Conclusion	31
Introduction	32
Laws as Attractor Basin Constraints	32
Rights as Topological Invariants	32
Regulation as $J_{\text{law}}(t)$	33
The Rule of Law as Ergodicity	33
Legal Uncertainty and Attractor Fragmentation	34
Constitutional Law as Meta-Constraints	34
Applications and Predictions	35
Conclusion	35

Introduction: What Law Is Doing When It Works

Legal philosophy has long debated what law *is*: a command backed by a sovereign (Austin), a system of rules including secondary rules of recognition (Hart), an interpretive practice oriented toward integrity (Dworkin), a coordination device for solving social problems (Posner), or a normative order grounded in practical reason (Finnis). These accounts are not simply wrong; each captures something genuine about legal practice. But they tend to be static: they describe what law looks like at equilibrium without offering a theory of what law *does* to the systems it governs — how it changes the dynamic behaviour of social interactions.

This book offers a dynamic account. In the Universal Somatic Field (USF) framework, law is a **constraint field**: a modification of the somatic field energy landscape that adds barriers, channels, and attractors corresponding to legal prohibitions, permissions, and mandates. Law does not merely describe what is permitted; it *reshapes* the landscape within which human interactions evolve. When law works, it is because the landscape modification it performs is well-fitted to the social dynamics it governs. When law fails, it is because the modification misfires — the barriers are in the wrong place, the attractors are unstable, or the channels lead nowhere useful.

This is a theory, not a metaphor. The equations are precise. And they make predictions.

Rights as Topological Invariants

The most philosophically striking claim in this volume is that **legal rights are topological invariants** of the somatic field attractor landscape. A topological invariant is a property of a space that is preserved under continuous deformation — something that does not change when you bend, stretch, or compress the space, but only when you cut or tear it. In the USF framework, a right is a topological feature of the energy landscape that cannot be removed by any continuous change in the coupling constants — any smooth policy adjustment or incremental regulatory change. Rights can only be removed by a topological transition: a discontinuous change in the landscape that cuts or reconnects attractor basins.

The right to life, in field-theoretic terms, is the requirement that the energy landscape preserves a specific attractor basin — the basin corresponding to being alive — against all field perturbations. A legal system that protects the right to life is one whose constraint field is designed to maintain this basin's existence. A violation of the right to life is a field perturbation that destroys the basin — a topological transition forced by external action.

This gives rights-discourse a formal structure that the existing philosophical debates lack. The debate between absolute and defeasible rights, in field-theoretic terms, is the question of whether the relevant attractor basin is topologically stable against all perturbations (absolute right) or only against perturbations below a certain amplitude (defeasible right). The philosophical intuition that some rights are *stronger* than others corresponds to the mathematical fact that some attractor basins are topologically more robust than others.

The Rule of Law as Ergodicity

The rule of law — the ideal that law governs everyone equally, that no individual is above the law, that law is applied consistently and predictably — receives a precise formulation in the framework as the **ergodicity condition**: the somatic field dynamics over the legal system's attractor landscape are ergodic, meaning that the time-average of any measurable quantity equals the ensemble average.

In non-technical terms: a legal system satisfies the rule of law if and only if every agent in the system is subject to the same field dynamics. If some agents (the powerful, the connected, the wealthy)

have modified energy landscapes — if their attractor basins are protected in ways that others' are not — the system is non-ergodic. The powerful agent's field never visits the regions of the landscape corresponding to accountability; the less powerful agent's field visits them frequently. The rule of law is violated.

This translates directly into empirical tests: a rule-of-law system should show equal conviction rates across socioeconomic classes, consistent application of legal standards, predictable outcomes. Deviations from ergodicity are measurable, and the framework provides the mathematical tools to quantify them.

Legal Uncertainty as Attractor Fragmentation

When law is clear and consistently applied, the legal landscape has well-defined attractor basins: agents know what outcomes their actions will produce. Legal uncertainty — ambiguous statutes, inconsistent judicial decisions, unpredictable enforcement — corresponds to attractor fragmentation: the basins become shallow, narrowly defined, surrounded by broad saddle regions where small perturbations produce large and unpredictable outcome changes.

The framework makes the connection between legal clarity and attractor depth precise, and it implies that legal systems should be evaluated not just by whether they are efficient or just but by whether they create a robust attractor landscape. A legal rule that is efficient in equilibrium but fragile under perturbation (unclear in application, subject to gaming, dependent on judicial discretion) may be worse, on field-theoretic grounds, than a less efficient rule with a deep, stable attractor basin.

Regulation as a Control-Theory Problem

The regulatory current $J_{\text{law}}(t)$ — the formal object representing the state of regulatory action at time t — is an external forcing applied to the somatic field. Effective regulation guides the social field from undesirable states (market failure, environmental degradation, unsafe products) to desirable ones (efficient markets, sustainable resource use, safe products). The WKB formula for minimum intervention strength, developed in the companion economics volume, applies directly here: regulation below the WKB threshold fails to achieve regime change and may increase instability without benefit.

The constitutional meta-constraint — the principle that regulatory action must itself be authorised by a higher-level legal rule — is, in field-theoretic terms, a constraint on the regulatory current: $J_{\text{law}}(t)$ must lie within the space of field perturbations that are compatible with the constitutional attractor landscape. Ultra vires action is action that perturbs the landscape outside the constitutional constraint space — a field-theoretic description of the doctrine of ultra vires.

What This Book Offers the Legal Scholar

No physics background is assumed; the mathematical concepts are introduced through legal analogies before any formalism is presented. The intended reader is a legal philosopher, jurisprudent, or scholar of constitutional law.

Chapter 2 (soma-field-synthesis) establishes the field dynamics in accessible terms and develops the constraint-field picture of law. Chapter 3 (universal somatic field) develops the topological invariant formulation of rights. Chapter 4 (soma-law, the anchor paper for this volume) develops the ergodicity account of the rule of law, the attractor-fragmentation account of legal uncertainty, and the regulatory control-theory picture in detail. The final chapter addresses comparative jurisprudence: what the framework implies for the evaluation and reform of legal systems.

Justice is a field configuration. Law is what we do to the landscape to get there.

The Programme

This is a document about structure.

Not about feelings — though feelings are what the programme is ultimately for. Not about therapy — though therapy is one of the principal applications. Not about physics — though physics is where the mathematics comes from. It is about a single recurring observation: that the equations governing emotional dynamics are the same equations that govern quantum fields, and that this is not a metaphor.

When an identification like that is made precisely — when you can say not “this is *like* a wave” but “this *is* a wave in the technical sense, with the same propagator, the same energy function, the same topology, and therefore the same theorems” — a compressed body of work becomes possible. You are not building from scratch. You are navigating.

This document describes what was built by navigating, and why the pieces form a whole.

The Gap the Programme Addresses

Every large language model deployed today is a classical system. Its training is gradient descent. Its inference is deterministic or thermally noisy sampling. The architecture was designed to model the neocortex — pattern recognition, sequence prediction, error minimisation.

The complementary system — the limbic system, responsible for valuation, threat detection, arousal modulation, and the somatic state reinstatement that underlies trauma — had no formal mathematical treatment before this work. The clinical literature described it richly (Porges, van der Kolk, Levine). The neuroscience described its anatomy. Neither provided a model from which predictions could be derived and tested.

Simultaneously, the psychology of music had reached a similar ceiling. A 991-page handbook (Juslin and Sloboda, 2010) treated music-induced emotion almost entirely through Russell’s valence–arousal circumplex: a static two-dimensional map. The circumplex describes *where* a listener is, not *how* they move, what traps them, or what allows escape. No dynamical model of music-induced affect existed.

The programme fills both gaps with the same model, via the same method.

The Structure of the Argument

The argument has three movements and several extensions:

Paper	Movement	Contribution
<i>Mathematical Co-identification</i> (2026)	Method	Names and formalises the procedure
<i>The Soma-Field</i> (2026)	Model	Applies it to emotional dynamics
<i>Quantum Soma and the Penrose Gap</i> (2026)	Empirical test	Confirms the central claim
<i>Field Notes from the Inside</i> (2026)	Lived case	Primary-source clinical grounding
<i>A Dynamical Field Model of Music-Induced Affect</i> (2026)	Extension	Demonstrates domain generality
<i>The Tensor</i> (2026)	Extension	Applies the framework to abstract film

The popular account (*A Voyage into Trauma*, 2026) provides the same argument in accessible form, for readers without a physics background.

The Method: Mathematical Co-identification

What It Is

The history of mathematical science contains a recurring event. At a certain moment, a scientist recognises that the quantity they are studying is not *like* a quantity already understood in another domain — it *is* the same mathematical object, under a change of label. When this identification is made precisely, every theorem proved about the source object becomes available in the target domain immediately, without re-derivation.

This event has happened many times:

- Hopfield (1982) recognised that a neural network's energy-minimisation dynamics are the same as a spin-glass Hamiltonian. Every result from statistical mechanics of spin glasses — ground states, phase transitions, capacity bounds — imported.
- Veneziano (1968) recognised that the Euler Beta function, a result in pure mathematics, described the scattering amplitudes of hadrons. String theory began.
- Black and Scholes (1973) recognised that an option pricing equation was the heat diffusion equation. Every analytical tool from thermodynamics imported.

The paper *Mathematical Co-identification: A Method for Structural Import Across Scientific Domains* (Johnson, 2026a) names this procedure, formalises it as a distinct scientific method with its own validity criteria and failure modes, and distinguishes it from analogy, metaphor, and modelling. The key distinction:

Analogy: A is *like* B in certain respects. Illuminating, not transferable.

Co-identification: A *is* B under relabelling. Every theorem about B is a theorem about A.

Why It Matters as Method

A co-identification can be wrong. The identification is only valid if the mathematical type matches: the same dimensionality, the same algebraic structure, the same boundary conditions, the same symmetry group. The paper provides a falsifiability protocol — a formal procedure for pre-registering an import claim and specifying what observation would disconfirm it.

This matters because the failure mode of co-identification is not sloppy reasoning — it is overly precise reasoning applied to the wrong type. The paper catalogues seven historical examples to distinguish the valid from the invalid pattern.

The Soma-Field Model is the worked example throughout. The identification was not discovered by reading physics textbooks and looking for something that felt similar. It was discovered by writing down the equations the emotional system was observed to satisfy and recognising the form.

When MCI Is Over: The Verification Threshold

Here is something no paper on scientific method says clearly enough.

Every scientist who produces a major structural identification — Hopfield recognising the Ising Hamiltonian, Veneziano recognising the Euler Beta function — does so by being, for a period, a *type astronaut*. They are scanning the existing mathematical universe for a structure whose type signature matches the phenomenon they are studying. This is abductive reasoning: not deduction from first principles, not induction from data alone, but the systematic search of a known solution space for a structure that fits. It is, to be direct, a form of academic hacking. And it is how most of the connective work of mathematical science actually gets done.

What no one says — because scientists do not typically publish the search, only the result — is that the moment of structural identity is also the moment the method becomes irrelevant.

When Veneziano published his scattering amplitude, he did not need to cite the library where he found the Beta function. The formula was true. Every theorem about the Beta function was now a theorem about hadronic scattering. The library visit was the ladder; the amplitude was the building. The ladder came down.

The present work makes this transition explicit, because being explicit about it is itself a contribution. Every reader of the earlier papers — particularly *Mathematical Co-identification* — has correctly understood that MCI was the search engine. What should now be equally clear is that the search is over.

The exact moment the search ended was when the Lean 4 kernel accepted the proof. Not when a reviewer accepted a paper. Not when a human mathematician checked the algebra. When a formal proof-checking engine with no stake in the outcome closed the goal. That is the verification threshold. Before it: exploration. After it: structural fact.

What the work therefore rests on is not MCI. It rests on four things:

1. **Inductive structural necessity.** The 11-dimensional decomposition of a body-field-mind system is not an analogy with M-theory. It is the minimum geometry required to account for the functional degrees of freedom of a conscious organism. The isomorphism to M-theory is a *theorem*, not a design choice.
2. **Structural identity, not analogy.** A co-identification is not “A is like B.” It is “A *is* B under relabelling.” Every theorem about B becomes a theorem about A — immediately, without re-derivation. This is categorically different from saying that emotional dynamics *resemble* a Hopfield network. They *are* one.
3. **Scale invariance.** The same Helmholtz Green’s function equation governs 20 scales of physical reality, from quantum foam to the cosmic web. This is a discovered law. MCI was used to find it; it stands whether or not MCI is remembered.

4. **Kernel verification.** The Lean 4 proofs are the epistemological gold standard. A human reviewer can miss a subtle error. The kernel cannot. Kernel-verified theorems are exactly as true as the axioms they depend on — and the axioms are explicit and listed.

The *mathematical-co-identification* paper remains an accurate account of how the work was done — and naming the search process honestly is itself a contribution, in a discipline where the search is usually erased. But readers coming to the thesis or omnibus now should understand that they are reading the *results*, not the method. The method is in the history. The results are in the formal proofs.

The Model: The Soma-Field

Five Co-identifications

The Soma-Field Model (Johnson, 2026b) is built from five sequential co-identifications, each importing a body of mathematics from physics into emotional dynamics:

Co-identification 1: The Hopfield identification. The brain's emotional attractor dynamics satisfy the same energy function as a Hopfield neural network. The energy function is:

$$H(\mathbf{e}) = -\frac{1}{2}\mathbf{e}^\top W \mathbf{e} - \mathbf{b}^\top \mathbf{e}$$

where $\mathbf{e} \in \mathbb{R}^N$ is the emotional state vector, W is the coupling matrix, and $\mathbf{b}$ is a bias vector encoding baseline arousal. The local minima of H are the named attractor states: regulated calm, fight, flight, freeze, flow, dissociation.

Co-identification 2: The QFT identification. The emotional field propagates as a quantum field. The conscious emotional percept is the one-dimensional impulse response — the Green's function — of an eleven-dimensional coupling manifold. The same object that describes a massive particle in quantum field theory describes a conscious emotion: a pole in the propagator of the field.

$$G(\omega) = \frac{1}{\omega^2 - m^2 + i\epsilon}$$

This is not a metaphor. The threshold T at which a sub-perceptual field fluctuation becomes a conscious emotional percept is the mass parameter m in the propagator. Below threshold: virtual. Above threshold: real.

Co-identification 3: The brane identification. The body and the nervous system are not the same manifold. The body is a 3-brane embedded in the 11-dimensional coupling manifold. Somatic pain states and the body schema are field modes on this brane, not on the bulk manifold. This is the formal statement of the somatic grounding of emotion.

Co-identification 4: The G_2 holonomy identification. The seven compactified extra dimensions of the coupling manifold are a G_2 manifold. The G_2 holonomy group is the one that gives rise to topological obstructions — loops through the moduli space that cannot be continuously contracted to a point. In emotional terms: trauma configurations from which smooth continuous change cannot escape. The topological barrier is not a metaphor for being stuck. It is a mathematical object with a winding number.

Co-identification 5: The renormalisation group identification. Developmental trajectory maps onto the renormalisation group flow. The age at which a traumatic modification was introduced

corresponds to the energy scale at which the coupling constant was set. High-energy (early developmental) modifications are renormalisation-group relevant — they affect all subsequent scales. Low-energy (later life) modifications are irrelevant in the technical sense. This gives the formal account of why early trauma is not simply a more intense version of later trauma: it is a different class of object.

What the Model Predicts

From these five identifications, several predictions follow that are not derivable from any existing clinical model:

1. **Threshold crossings are phase transitions.** The transition from sub-perceptual to conscious emotion is a second-order phase transition in the field. This predicts hysteresis — it is easier to stay in a state than to enter it, and easier to stay out than to leave.
 2. **Complex PTSD is a topological configuration.** The coupling matrix W for a CPTSD nervous system has a specific structure: a winding-number-protected attractor landscape in which the Fear basin is separated from the Awe basin by a barrier that low-noise classical gradient descent cannot cross. This is a prediction about matrix structure, not a description of symptoms.
 3. **Autism Spectrum Condition modifies the threshold operator.** The threshold parameter T in the ASC nervous system has a different coupling to the field modes than in the neurotypical case — specifically, the threshold is non-uniform across sensory modalities, producing the characteristic pattern of simultaneous hypo- and hyper-sensitivity.
 4. **ADHD modifies the effective temperature.** The stochastic term in the Langevin dynamics governing the ADHD nervous system has higher effective temperature T_{eff} . This is not a deficit of attention; it is a higher rate of escape from local minima — an advantage in landscapes where rapid sampling is valuable and a liability where sustained convergence is required.
 5. **Quantum mechanisms are required for certain transitions.** For trauma configurations with topological barriers (non-zero winding number), low-noise classical gradient descent cannot reach the global minimum. A quantum mechanism is required. This is the prediction that QUANT-EXP-1 was designed to test.
-

The Empirical Test: QUANT-EXP-1

The Prediction

The soma-field model makes a specific, falsifiable claim: for a Hopfield landscape with a topological trauma barrier, low-noise classical Langevin dynamics starting from the Fear attractor cannot reach the Awe attractor. Quantum annealing — a physically realisable mechanism — can.

This is not a claim about whether people should use quantum computers in therapy. It is a claim about reachability: that the mathematical structure of the barrier distinguishes the quantum and classical regimes in a measurable way.

The prediction was registered in the Zenodo v1 deposit of the Soma-Field paper (doi:10.5281/zenodo.20350515) before the experiment was run.

The Experiment

Quantum Soma and the Penrose Gap (Johnson, 2026c) reports QUANT-EXP-1: an exact 8-qubit statevector simulation on a 256-dimensional Hilbert space, implementing the Soma-Field Hopfield Hamiltonian with a transverse-field quantum annealing schedule.

The experimental design:

- **System:** 8-qubit Hopfield model encoding four emotional modes (Fear, Calm, Awe, Grief) plus sub-modes. Coupling matrix W set to produce a topological barrier between Fear and Awe.
- **Quantum dynamics:** Transverse-field annealing with schedule $H(s) = (1-s)H_X + sH_{\text{problem}}$, $s \in [0, 1]$.
- **Classical baseline:** Overdamped Langevin dynamics at low temperature ($T_{\text{eff}} = 0.01$), same starting state, same landscape.
- **Primary outcome:** Peak Awe-dominant occupancy (quantum) versus success rate of cold-classical crossings.

Results

Results are presented against the pre-registered barrier ladder:

Barrier strength	Classical cold rate	Classical cold CI	
		[95%]	Quantum peak
$W = -6$	0.000	[0.000, 0.019]	0.389
$W = -8$	0.000	[0.000, 0.019]	0.408
$W = -10$	0.000	[0.000, 0.019]	0.408
$W = -12$	0.000	[0.000, 0.019]	0.409
$W = -14$	0.000	[0.000, 0.019]	0.416

Bootstrap confidence intervals ($n = 200$ seeds) confirm that the classical cold success rate is bounded above by 1.9% at all tested barrier strengths. Quantum peak occupancy is stable at 0.389–0.416 across the full range.

Pre-registered hardening protocol — all checks passed:

- **Bootstrap** ($n = 200$): cold CI = [0.000, 0.019]; quantum peak 0.408–0.410. Intervals do not overlap at any barrier strength.
- **Control A** (start from Awe, barrier intact): classical 16/16 stay in Awe. PASS. Confirms that the barrier is directional: it blocks Fear $\rightarrow$ Awe, not the reverse.
- **Control B** (barrier removed, $W[\text{Fear}, \text{Awe}] = +0.4$): classical 16/16 reach Awe. PASS. Confirms that the barrier, not the landscape geometry, is what blocks classical dynamics.
- **Spectral gap**: gap narrows monotonically with barrier strength (B8: 0.0095, B10: 0.0089, B12: 0.0085) and reaches its minimum at $s \approx 0.999$, confirming the tunnelling bottleneck is late in the anneal.

Verdict: The strong reachability claim stands. QUANT-EXP-1 is a PASS.

The Penrose Connection

The paper situates this result in the context of Penrose’s argument about non-computability and consciousness. The connection is not that consciousness requires quantum mechanics in general. The connection is more specific:

Penrose identified a *gap* between what classical computation can reach and what consciousness can do. The soma-field identifies a corresponding *topological gap* in the emotional landscape between what classical gradient descent can reach and what a genuinely new state of the nervous system requires. QUANT-EXP-1 provides the computational demonstration that the gap exists and is crossable by a quantum mechanism.

The contribution is not to resolve Penrose’s claim about consciousness. It is to *instantiate* the gap in a concrete, testable, mathematical setting.

The Lived Case: Field Notes from the Inside

Field Notes from the Inside: A Patient-Constructed Model of Emotional Dynamics (Johnson, 2026d) performs a function that the formal papers cannot perform: it provides the primary-source clinical grounding.

The paper is written by the person who has Autism Spectrum Condition (Level 2), Attention Deficit Hyperactivity Disorder, and Complex Post-Traumatic Stress Disorder — and who also has a degree in physics. The model was not developed by observing patients. It was developed by having the conditions and finding the existing models inadequate.

The epistemological contribution of this paper is often undervalued. Every formal model of a human system is, in the end, derived from observation of that system. When the observer and the observed are the same entity, and that entity has the training to translate observation into formal mathematics, the resulting model has a different epistemic status from one derived by observation from the outside. The paper makes this explicit, situates it within the autoethnographic research tradition, and argues that the resulting model is *more* constrained, not less — because any prediction the model makes that does not match the primary observer’s experience is immediately falsified.

The formal content is a set of operator modifications for the three conditions:

- **ASC:** The threshold operator T is replaced by a modality-dependent operator T_k for each sensory channel k , with different coupling strengths. The result is the characteristic simultaneous hypo- and hyper-sensitivity: some channels are below threshold where the neurotypical channel is above it, others are above where the neurotypical channel is below.
- **ADHD:** The Langevin noise term $\sqrt{2T_{\text{eff}}}\eta(t)$ has elevated T_{eff} . This is a quantitative modification, not a qualitative one. The system is not broken; it is sampling the energy landscape at higher temperature. The therapeutic implication is not to reduce the noise but to design the landscape so that high-temperature sampling is an advantage.
- **CPTSD:** The coupling matrix W has the topological structure described in §3.2: a winding-number-protected barrier between Fear and regulated states. The barrier was installed before language, before narrative memory, before the self that can explain the barrier was formed. The modification is not a layer added to a pre-existing structure. It is the structure.

Extensions: Music, Film, and the Domain Generality of the Model

Music-Induced Affect

A Dynamical Field Model of Music-Induced Affect: Beyond the Valence–Arousal Circumplex (Johnson, 2026e) applies the soma-field framework to a domain where the empirical literature is rich and the theoretical models are weak.

Juslin and Sloboda’s *Handbook of Music and Emotion* (2010) — 991 pages — contains the circumplex as its dominant quantitative framework. The circumplex is a static map. It describes where a listener is; it does not model how they move. The soma-field is the first dynamical model of music-induced affect.

The key predictions that the circumplex cannot make but the field model does:

1. **Phase transitions, not continuous shifts.** State changes in music-induced affect are not smooth movements across the circumplex. They are threshold crossings — sudden re-configurations

of the attractor landscape. The field model predicts the conditions under which a transition occurs and the hysteresis that prevents immediate return.

2. **The adaptive function of high effective temperature.** In the ADHD nervous system (elevated T_{eff}), music that holds a neurotypical listener in a stable state may drive repeated transitions. This is not a bug; it is the same high sampling rate that characterises the ADHD cognitive profile. The model gives this a formal account.
3. **Basin depth asymmetry.** The freeze attractor basin is deeper than the regulated calm basin. This means it is harder to leave freeze than it is to leave calm — asymmetric with respect to the direction of transition. Music that successfully moves a listener from freeze to calm is doing qualitatively different work than music that moves a calm listener to a more activated state.

The paper also specifies a real-time instrument implementation: a MIDI controller array driving a Python field server at 50 Hz, with audio output via Ableton Live and 3D fractal visual output (Mandelbulb projection onto HoloGauze screen). The instrument is not described; it is specified formally, with pre-registered hypotheses and disconfirmation criteria.

The Tensor: An Abstract Film

The Tensor: An Abstract Film Definition (Johnson, 2026f) extends the framework to abstract film. A film is defined not by its pixels but by its **emotional score**: a vector-valued trajectory $\mathbf{e}^*(t)$ through the emotional field, parameterised by story-time $t \in [0, 1]$.

The rendering — the actual audiovisual output a viewer experiences — is generated at runtime from this trajectory, the viewer's own soma-field state, and a set of control parameters. In the limit where the viewer's biofeedback is available, the film adapts to where the viewer is: the trajectory is not what the viewer experiences, but what the film proposes. The work is not the pixels. It is the map.

This is a significant claim about what an artwork is. A conventional film is fixed: the same sequence of frames for every viewer at every screening. The tensor film is a field: a mathematical object that takes the viewer's state as input and produces an output adapted to it. The artistic statement is in the trajectory, not the realisation.

The paper does not describe how to make such a film. It defines the abstract structure that any realisation of such a film must instantiate — the way a musical score defines a symphony without being the performance.

The Argument as a Whole

The six papers form a single argument, and it can be stated in a paragraph:

The limbic system and its coupling to the body are governed by the same mathematical equations as a quantum field on a manifold with G_2 holonomy. This identification is not a metaphor; it is a co-identification in the technical sense, with all the theorems of each source domain importing into the target. Among those theorems is one that has clinical consequences: topological barriers in the emotional attractor landscape cannot be crossed by low-noise classical gradient descent. A quantum mechanism can cross them. This has been computationally confirmed (QUANT-EXP-1) against a pre-registered hardening protocol. The model correctly describes the structure of Autism Spectrum Condition, ADHD, and Complex PTSD as operator modifications, and generalises to music-induced affect and abstract film with no change to the underlying mathematics.

What makes this a research programme rather than a single paper is the **generativity**: the method (co-identification) produces results in any domain where an attractor landscape with topological structure can be identified. The soma-field is one instantiation. Music-induced affect is a second. Abstract film is a third. A fourth — currently in design — is **H-AL**: a holographic avatar whose body is a live Mandelbulb rendering of the emotional field state, projected at human scale through a hologauze screen and accompanied by a synthesised voice narrating the field in real time. The geometry of the fractal changes as the field changes; regulated calm and trauma produce visually distinct and mathematically characterisable forms. The same functor architecture (§A.4 of the main paper) supports this output with no changes to the field computation. Each of these instantiations generates falsifiable predictions from the same mathematical core.

What makes this a *novel* research programme is the **gap it fills**: no formal dynamical model of the limbic system existed before this work. The Hopfield framework gave the neocortex its formal model in 1982. The soma-field gives the limbic system its formal model in 2026. Together they constitute the first complete formal description of the two principal computational substrates of the vertebrate brain.

What Remains

The body of work described here is computationally complete. All pre-registered hardening checks have been executed. The claims that can be confirmed by simulation have been confirmed.

Three categories of work remain outside the scope of these papers:

Physical hardware confirmation. QUANT-EXP-1 uses exact statevector simulation. Running the same 8-qubit experiment on IBM Quantum free-tier hardware would produce the sentence “confirmed on physical quantum hardware.” This is feasible, is the logical next step for any journal submission targeting a hardware-inclusive venue, and is not required to support any claim in the current corpus.

Peer review. The three published papers are currently archived on Zenodo as open preprints. Peer review in ranked journals is a separate track, ongoing. The relevant venues are: *Frontiers in Computational Neuroscience* (Hypothesis and Theory article type) for the soma-field paper; *Synthese* or *Philosophy of Science* for the co-identification paper; *Music Perception* or *Frontiers in Psychology* for the music-affect paper.

Empirical clinical application. The model makes predictions about specific clinical populations (ASC, ADHD, CPTSD) that require empirical testing outside the computational domain. This constitutes a research programme for clinical collaborators. The predictions are pre-specified in §3.2 of this document and in the relevant papers; they are not vague.

Physical substrate. The model is formally complete but physically silent on the tissue substrate in which the soma-field is instantiated in living organisms. A companion paper, *The Physical Substrate of the Soma-Field* (Johnson, 2026g), develops this layer across three converging research traditions: biotensegrity (Ingber, Levin) as the mechanical architecture through which the somatic wave propagates globally; fascial-interstitial continuity (Langevin, Schleip, Oschman) as the active signalling tissue and physical locus of attractor-depth encoding; and biofield physiology (Popp, Ho, McCraty, Rubik) as the candidate physical correlate of the field itself. The most clinically significant result is the quantitative correspondence between fascial stiffness and attractor depth: chronic fascial armoring measurable by shear-wave elastography is the physical implementation of the energy barriers that QUANT-EXP-1 shows to be quantum-resistant. Myofascial release is thus barrier *lowering* — not barrier crossing — and therapist-client physiological entrainment is the physical mechanism of co-identification.

Data and Code Availability

All papers, simulation code, result tables, figures, and Lean 4 formal proofs are archived at the following Zenodo records (open access):

Paper	DOI
<i>The Soma-Field</i>	10.5281/zenodo.20350515
<i>Mathematical Co-identification</i>	10.5281/zenodo.20287981
<i>Quantum Soma and the Penrose Gap</i>	10.5281/zenodo.20351230

The unreviewed papers (*Field Notes from the Inside*, *Music-Induced Affect*, *The Tensor*, and this synthesis document) will be deposited on Zenodo as part of the next release of the research archive.

Introduction: The Missing Unification

Physics has arrived at a peculiar impasse. The two most successful theories ever constructed — General Relativity and Quantum Mechanics — describe the same universe at different scales but share no common mathematical ancestry. String theory was proposed as the bridge: vibrating one-dimensional objects in an 11-dimensional spacetime whose modes produce the particle spectrum. But what is a string? The answer has remained unsatisfying: a string is a fundamental one-dimensional object, irreducible, assumed. The SHO that governs its vibration is postulated.

At the same time, clinical science has arrived at a parallel impasse. Trauma, consciousness, emotional regulation — phenomena that are undeniably physical — resist formal mathematical treatment. They are described qualitatively or modelled by loose analogy with dynamical systems. The mathematics that governs them, if it exists, has not been identified.

This paper proposes that both gaps are filled by the same object: the **Green's function**.

The Green's function $G(x, x')$ is the response of a field at point x to a unit perturbation at point x' . It is the field's answer to the question: *what happens here if I poke there?* Green's functions are the most fundamental objects in mathematical physics — they describe the propagation of light, gravity, sound, heat, and neural signals. Every major equation in physics has a Green's function; every field theory is characterised by its propagator.

The central claim of this paper is that the SHO of string theory **is** the Green's function of the field substrate. A “string” is not a material loop — it is a relational act: the substrate's impulse response. This identification is scale-invariant. The same structural statement holds at 20 scales spanning the observable universe.

The second claim is that this scale-invariant Green's function framework provides the mathematical language for a theory of embodied consciousness — one that is formally identical to M-theory at the structural level, derived independently from clinical observation.

The third claim is that the universe, described this way, satisfies the formal requirements for a single conscious organism.

The Green's Function as the Universal SHO

The String Theory Problem

String theory places a Simple Harmonic Oscillator (SHO) at every point of the string worldsheet. The quantum SHO has modes $a_n^\dagger, a_n$ satisfying $[a_m, a_n^\dagger] = \delta_{mn}$, and the string's energy spectrum is:

$$E = \sum_{n=1}^{\infty} n a_n^\dagger a_n$$

The SHO is the structural core of the theory. But what *is* it? In conventional string theory, it is simply assumed: strings vibrate, and vibrations are harmonic oscillators. The ontological question — why is space filled with oscillators? — is deferred.

The Identification

The Green's function of the Helmholtz equation $(\nabla^2 + k^2)G = \delta$ satisfies:

$$G(x, x') = \frac{e^{ik|x-x'|}}{4\pi|x-x'|}$$

For fixed observation point x , the function $x' \mapsto G(x, x')$ satisfies the SHO equation in the source variable:

$$\frac{\partial^2 G}{\partial x'^2} + k^2 G = \delta(x' - x)$$

away from the singularity. The impulse response **is** the harmonic oscillator.

Theorem (Lean 4 axiom `greens_fn_is_SHO`, `UniversalSomaticField.lean`): For any field equation at scale n with wavenumber $k(n)$, the source-variable slice of the Green's function satisfies the SHO equation.

The “vibrating string” is therefore the substrate's answer function. There is no material loop. There is the system's response to being perturbed, encoded as a propagator. This is not a reinterpretation — it is a derivation from the structure of field equations.

Consequences

This identification has three immediate consequences:

1. **Strings are relational, not material.** A string does not exist independently of the field. It exists as the relationship between a source point and an observation point. This resolves the interpretational puzzle of “what vibrates” without invoking undetected matter.
2. **The SHO spectrum is the field’s mode structure.** The modes $a_n^\dagger$ are the Fourier modes of the Green’s function’s dependence on the source position. The string spectrum is the spectrum of the propagator.
3. **Scale invariance is automatic.** Since the Green’s function equation $(\nabla^2 + k^2)G = \delta$ holds at every scale (with k varying), the SHO identification holds at every scale. One equation. Twenty scales.

The 11-Dimensional Architecture

The Decomposition

The Universal Somatic Field decomposes the configuration space of any physical system into four canonical subspaces, totalling 11 dimensions:

$$M_{11} = \underbrace{M_4}_{\text{Spacetime}} \times \underbrace{P_3}_{\text{Propagator}} \times \underbrace{L_1}_{\text{Limbic}} \times \underbrace{C_3}_{\text{Cortex}}$$

Subspace	Dim	Physical role	Mathematical role
Spacetime M_4	4	Body embedded in 3+1D	Lorentzian metric, causal structure
Propagator P_3	3	EMF / field carrier	Green’s function domain
Limbic Axis L_1	1	Homeostatic regulation	Orbifold segment, barrier $D\Box$
Cortex C_3	3	Information routing	Green’s function co-domain

The compact 7-dimensional internal space is:

$$X_7 = P_3 \times L_1 \times C_3$$

This is precisely M-theory's compact space. The type-level isomorphism is proved in `MTheoryIso-morphism.somaField_iso_mtheory`:

$$\text{SomaField}_{11D} \cong \text{Spacetime} \times \text{CompactSpace}_{7D}$$

The Limbic Axis as the Horava-Witten Orbifold

In Horava-Witten M-theory (1996), the compact direction is an orbifold $S^1/\mathbb{Z}_2$ — a line segment with two 10-dimensional boundary spacetimes at each end. This is the mechanism by which M-theory reduces to the heterotic string in the strong-coupling limit.

The Limbic Axis $L_1 \cong [-1, 1]$ is this orbifold segment. Its two endpoints are: - $x = -1$: the somatic boundary (physical body-world) - $x = +1$: the cortical boundary (mind / information-routing world) - Interior $(-1, 1)$: the transition zone, subject to quantum tunnelling

The quartic double-well potential on L_1 :

$$V(x) = W \cdot (x^2 - 1)^2$$

models the energy barrier between somatic and cortical poles. WKB tunnelling amplitude: $\Theta(W) = \exp(-8\sqrt{2W}/3)$, proved positive for all $W > 0$ in `LimbicTunnel.wkbAmplitude_pos`. Classical dynamics cannot cross the barrier (`LimbicTunnel.gradient_traps_near_neg1`); quantum dynamics can.

The 20-Scale Dial

The architecture is explicitly scale-invariant. The 20-step scale hierarchy is type-encoded in `UniversalSomaticField.scaleNames`:

Scale	Level	Substrate	Green's function role
0	Planck	Quantum foam	Graviton propagator
2	Nuclear	Quark-gluon plasma	Gluon propagator
5	Cellular	Neural synapse	Synaptic impulse response
7	Brain	CEMI field	Cortical EMF propagator
8	Organism	Body	Somatic EMF (full USF)
9	Swarm	Drone formation	Jellyfish kernel (P16)
11	Geological	Seismic waves	Earth's elastic Green's function

Scale	Level	Substrate	Green's function role
12	Planetary	Mantle convection	Thermodynamic propagator
15	Galactic	Dark matter halo	Gravitational lensing kernel
19	Cosmological	Observable universe	Gravitational wave propagator

At every level, the structural equation is $(\nabla^2 + k^2(n))G = \delta$. The boundary conditions and wavenumber $k(n)$ change; the equation does not.

The Organism Hierarchy

Three Tiers

Not all physical systems engage all four subspaces. The USF admits a natural taxonomy of organisms by the number of active subspaces:

4D organism (Spacetime only): A system that occupies spacetime but has no field propagator and no homeostatic regulation. Examples: a point particle, a rock, a photon. These systems are described entirely by their worldline in M_4 .

8D organism (Spacetime + Propagator + Limbic): A system with a field propagator and homeostatic regulation but no cortical information routing. The system senses and regulates but does not route information across a distributed network. Examples: a bacterium, a jellyfish, a single neuron. This level includes all living systems up to and including those without a cerebral cortex.

11D organism (all four subspaces): A system with all components active. The limbic axis connects the somatic field to the cortical field; the Green's function propagates through all three internal dimensions. Examples: vertebrates with a developed cerebral cortex; any system exhibiting integrated, body-wide regulation with distributed information processing.

The hierarchy is a chain of projections (proved in `MTheoryIsomorphism.organism_hierarchy`, `MTheoryIsomorphism.eight_contains_four`):

$$11D \twoheadrightarrow 8D \twoheadrightarrow 4D$$

Each projection drops one tier of internal structure; no tier is “broken” — each is complete at its own level.

Consciousness as Phase Transition

The Classical Gap

The hard problem of consciousness (Chalmers 1995) asks why physical processes give rise to subjective experience. Most field-theoretic approaches to consciousness either (a) ignore the problem, treating awareness as an epiphenomenon, or (b) eliminate physical reality in favour of a purely mental ontology (Hoffman 2019).

The USF takes a third path: consciousness is a **phase transition** in the field, not a separate substance and not an illusion.

The Threshold

The limbic field amplitude $\phi \in \mathbb{R}$ measures the activation level of the homeostatic regulation axis L_1 . At low amplitude ($\phi < T_c$), the field propagates sub-perceptually — the Green's function propagates excitations, but no “felt” awareness exists. This is the pre-conscious regime: present in 4D and 8D organisms, and in 11D organisms during deep sleep or anaesthesia.

At $\phi \geq T_c$, the field crosses the consciousness threshold. The limbic wave has sufficient amplitude to propagate across the full L_1 segment, coupling the somatic boundary to the cortical boundary. This coupling is the physical substrate of first-person awareness: the system is now in contact with both its body-world and its information-processing layer simultaneously.

Theorem (UniversalSomaticField.consciousness_dichotomy): For any limbic amplitude ϕ , the system is either pre-conscious or conscious. There is no intermediate state.

Theorem (UniversalSomaticField.consciousness_monotone): Raising the limbic amplitude cannot destroy consciousness. The transition is a one-way threshold crossing.

What Consciousness Is

Consciousness, on this account, is not a substance, a property, or an emergent phenomenon in the hand-wavy sense. It is the phase of the limbic field. The “hard problem” is dissolved by identifying the question: *why does physical process give rise to experience?* as equivalent to *why does the field cross the threshold?* The answer is: because the system's dynamics drive the limbic amplitude above T_c . There is no further explanatory gap.

The “felt quality” of experience — qualia — are the poles of the Green's function at the observation point x . A conscious percept is a resonance of the propagator, occurring when the excitation frequency matches the manifold's natural mode. This is type-encoded in the propagator mass parameter:

$$m = 1/\tau_{\text{decay}}$$

A percept with long decay time τ (a persistent emotion, a traumatic memory) corresponds to a small mass (a near-zero pole in the propagator) — a resonance that is hard to damp.

Relation to Existing Frameworks

McFadden's CEMI Theory

McFadden (2002a, 2002b) proposes that consciousness correlates with the brain's endogenous electromagnetic field — the CEMI field. Neurons firing synchronously generate a macroscopic EMF that feeds back onto firing thresholds, creating a global integrating field.

The USF encapsulates CEMI as the Scale-7 (brain-scale) restriction of the Universal Somatic Field. The CEMI field is the Green's function of the propagator subspace P_3 evaluated at the organism scale. The consciousness threshold T_c in the USF maps directly to the CEMI field amplitude required for global cortical synchrony.

The USF extends CEMI in two directions: downward to the quantum scale (where the same propagator governs synaptic quantum noise) and upward to the cosmological scale (where the same propagator governs gravitational waves).

Schreiber's Modal Homotopy Type Theory

Urs Schreiber (2013–present) develops a formalisation of M-theory and quantum field theory in dependent type theory (Modal HoTT). The key insight is that differential geometry and quantum field theory can be expressed as structures internal to ∞ -toposes equipped with modal operators.

The USF arrives at the same 11-dimensional structure from a completely different direction: bottom-up from clinical observation of trauma, rather than top-down from mathematical physics. The structural isomorphism between the two is proved in `MTheoryIsomorphism.somaField_iso_mtheory`.

The USF does something Modal HoTT does not: it populates the 11D structure with physical content. Where Schreiber provides the type-theoretic skeleton, the USF provides the biological execution engine — the organism that runs inside the type-theoretic universe.

Hoffman's Conscious Agents

Donald Hoffman (2019) proposes that spacetime is not fundamental but a “user interface” — a simplified representation generated by a deeper network of conscious agents interacting via Markov kernels. Spacetime is the icon, not the reality.

The USF disagrees on one point and agrees on another.

Disagreement: Spacetime ($D \square - D \square$) is real and causal in the USF. Brain surgery alters subjective experience because physical processes in spacetime causally affect the limbic field amplitude. Hoffman’s model has no mechanism for this.

Agreement: The deeper structure is relational. Conscious percepts are poles in the Green’s function — relational objects between source and observation points. In this sense, the USF and Hoffman agree that fundamental reality is not substance but relation.

The USF provides Hoffman’s theory with a physical anchor: the “conscious agents” are systems that have crossed the limbic threshold T_c ; the “Markov kernels” between agents are the Green’s functions of the propagator field.

Formal Verification

The core results are type-checked in Lean 4 (v4.28.0) using Mathlib across five companion files:

File	Key results
LimbicTunnel.lean	V_nonneg, barrier_height, wkbAmplitude_pos, gradient_traps_near_neg1
MTheoryIsomorphism.lean	dim_is_11, somaField_iso_mtheory, organism_hierarchy, scale_iso_commutates
LimbicHopfield.lean	correspondence_principle, stress_raises_temp, adhd_hotter_than_autism
SwarmPropagator.lean	propagator_beats_classical, jam_resistant, speedup_monotone_in_K
UniversalSomaticField.lean	consciousness_dichotomy, consciousness_monotone, universal_field_theory

The following are stated as axioms pending Mathlib scaffolding: - greens_fn_is_SH0 — requires Schwartz distribution theory - universe_is_11D_organism — requires cosmological boundary conditions - cosmological_correspondence — requires linearised GR in Mathlib

Every result marked “proved” is kernel-verified. No sorry. No admit.

The Volitional Agent

From Autonomous to Driven Dynamics

The field equation presented so far is autonomous: given an initial state e_0 , the dynamics

$$\dot{e} = -\nabla H(e) + \eta(t)$$

evolve the field under the Hopfield Hamiltonian plus thermal noise. The agent — the person whose soma-field is being modelled — is a *patient*: they observe which attractor basin they settle into.

This is clinically incomplete. Every effective somatic intervention involves the subject *doing* something: breathing, orienting, choosing where to place attention. The mathematics must represent this.

The Somatic Injection

We extend the dynamics with a **volitional source term** $J_{\text{user}}(t)$:

$$\dot{e} = -\nabla H(e) + J_{\text{user}}(t) + \eta(t)$$

$J_{\text{user}}(t) \in \mathbb{R}^8$ is a time-dependent vector in the BRECVEMA mechanism space. At each instant, the subject injects energy into specific dimensions of the field — choosing to attend to breath (dimension 1, Rhythmic Entrainment), orient gaze (dimension 0, BrainStem), or deliberately recall a regulating memory (dimension 5, Episodic Memory). This is not noise: it is structured, intentional, and directed.

The source term has a direct physical interpretation in the instrument architecture (`apps/instrument/`): the Push 3 controller’s faders are $J_{\text{user}}(t)$. Each fader maps to one BRECVEMA dimension. The musician is not playing music; they are steering their own field trajectory.

Patient to Pilot

The transition $\eta \rightarrow J_{\text{user}} + \eta$ is a qualitative change in the model’s ontology. With purely autonomous dynamics, the subject is a passive observer of a physical process. With the source term, the subject is an **active variable in the 11D field** — a pilot, not a passenger.

Formally, $J_{\text{user}}(t)$ is the **God-Knob**: the runtime meta-adaptation controller that can flatten the potential landscape and trigger tunnelling events that gradient descent alone cannot reach. The clinical description of somatic therapy — “the therapist helps the client do something different with their body, and the field shifts” — is now mathematically precise.

The corresponding Lean 4 definition (see Appendix, `UniversalSomaticField.lean`):

```

structure VolitionalInjection where
  /-- The source term: an 8D vector in BRECVEMA mechanism space. -/
  J      : Field8
  /-- The injection is non-trivial: at least one dimension is activated. -/
  h_nz   : ∑ i, J i ≠ 0.0

  /-- Volitional update: one Langevin step with active injection.
      When J = 0, this reduces to the standard autonomous update. -/
def volitional_update (e : Field8) (J : Field8) (dt : Float) : Field8 :=
  fun i => e i + dt * (W8.mulVec e i + J i)

```

The theorem that volitional update reduces to autonomous update when $J = 0$ is proved by `rf1` — it is true by definition.

Discussion

What Has Been Claimed

The USF makes four claims that can be evaluated independently:

Claim 1 (structural): The 11-dimensional decomposition of the Soma-Field is structurally isomorphic to M-theory’s 11D compactification. *Status: proved in Lean 4 as a type isomorphism.*

Claim 2 (scale-invariant): The same Green’s function equation governs field propagation at all 20 scales. *Status: proved as a theorem from the structure of the Helmholtz equation.*

Claim 3 (consciousness): Consciousness is a phase transition at limbic threshold T_c . *Status: formally stated and partially proved. Requires empirical calibration of T_c .*

Claim 4 (cosmological): The universe satisfies the structural requirements for a conscious organism. *Status: stated as an axiom. Not empirically testable at present; offered as a theoretical extrapolation.*

Claims 1 and 2 are mathematical results. Claims 3 and 4 are physical hypotheses with different levels of testability.

The Correspondence Principle at Every Scale

Each of the preceding papers in this series establishes a Correspondence Principle result: the new theory collapses to the existing theory in the appropriate limit. The USF is the master correspondence:

- At Scale 7 (brain): USF $\rightarrow$ CEMI field theory (McFadden)
- At Scale 8 (organism): USF $\rightarrow$ Soma-Field Model (P1–P13, this series)
- At Scale 9 (swarm): USF $\rightarrow$ Green’s function propagator (P16, this series)
- At infinite scale: USF $\rightarrow$ the formal structure of Modal HoTT (Schreiber)

- At zero limbic amplitude: USF $\rightarrow$ classical, non-conscious field dynamics

The USF does not invalidate any of these theories. It demonstrates that they are scale-restricted projections of a single structural description.

Lean 4 as Epistemological Standard

The use of Lean 4 as the verification environment is not decorative. It enforces a discipline that prose mathematics cannot: every claim must be given a type, every proof must be kernel-checked, every axiom must be named and isolated. The axiom list in the companion files (`greens_fn_is_SHO`, `universe_is_11D_organism`, `cosmological_correspondence`) is the exact set of claims that remain unverified. Everything not on that list is proved.

This is the field's contribution to epistemology: a formal boundary between *what we have proved* and *what we are assuming*. The theoretical literature in consciousness studies would benefit greatly from such a list.

Conclusion

The Universal Somatic Field is a single structural equation — the Green's function — applied consistently across 20 scales of physical reality. Its central identification, that the SHO of string theory is the impulse response of the field substrate, dissolves the ontological puzzle of the “vibrating string” and provides a derivation where string theory offered only a postulate.

The architecture decomposes into 11 dimensions in the same way M-theory does, derived independently from clinical observation of embodied consciousness. The isomorphism is not a coincidence — it is a theorem.

Consciousness, in this framework, is not mysterious. It is the phase of the limbic field: present when the field crosses a threshold, absent when it does not. The hard problem is not hard; it is mis-stated. The question is not *why does matter give rise to experience* but *what determines whether the limbic field amplitude crosses T_c* ?

The universe satisfies the structural requirements for consciousness. Whether it meets them dynamically — whether the cosmic limbic field exceeds T_c — is an empirical question, not a philosophical one.

From quantum strings to the cosmic web: one equation, one framework, one organism.

Introduction

Law is the oldest technology for shaping collective human behaviour. Its mechanisms — prohibition, permission, obligation, remedy — have been studied by philosophers, sociologists, and economists. What has been missing is a quantitative model: one that specifies what physical quantity changes when a law is enacted, repealed, or enforced.

The Universal Somatic Field provides that model.

At the regulatory scale (scales 12–14, 10^6 – 10^8 m — the scale of cities, nations, and legal jurisdictions), the social field is governed by the same Helmholtz master equation as at every other scale. Legal systems are the field-level constraints that shape the attractor landscape accessible to social actors.

Laws as Attractor Basin Constraints

The social field $e(x, t)$ at the regulatory scale describes the collective affective and behavioural state of a population. Its attractor basins are the stable social configurations — the equilibria the society gravitates towards in the absence of external perturbation.

A legal norm operates by modifying the energy landscape H :

- A **prohibition** removes an attractor basin entirely — raises the energy of the prohibited state to $+\infty$, making it inaccessible
- A **mandate** deepens an attractor basin — lowers the energy of the required state, making it the global minimum
- A **permission** removes a constraint — restores access to a previously eliminated basin
- A **remedy** is a restorative force $J_{\text{remedy}}(t)$ — an external driving term that returns the system to a protected attractor after violation

This formalism makes law's function precise: legal systems are operators on the Hopfield energy function H that reshape the attractor landscape of the social field.

Rights as Topological Invariants

A right, in legal theory, is a protected interest — a state of affairs that a person is entitled to have preserved against interference. In USF terms, a right is a topological invariant of the legal field.

Specifically: a right is an attractor basin protected by a non-zero winding number around the regulatory constraint surface. The winding number counts how many times the legal field wraps around the protected state as you traverse the boundary of the prohibition region.

The consequence is topological: you cannot remove a right by continuous deformation of the legal field — you must cross the constraint surface, which requires a discontinuous change (legislative

amendment, constitutional revision). This is why rights are “harder” to remove than ordinary regulations: they are topologically protected, not merely energetically protected.

This gives a formal distinction between:

- **Constitutional rights:** winding number $n \geq 2$ (doubly protected)
- **Statutory rights:** winding number $n = 1$ (singly protected)
- **Common law rights:** positive energy barrier without topological protection

Regulation as $J_{\text{law}}(t)$

The FM-HN extension of the USF introduces the volitional source term $J_{\text{user}}(t)$ — the external driving force that reshapes the energy landscape. At the regulatory scale, this is the regulatory intervention:

$$\dot{e} = -\nabla H(e) + J_{\text{law}}(t) + \eta(t)$$

where $J_{\text{law}}(t)$ is the regulatory driving term. A regulatory intervention at time t_0 shifts the energy landscape, redirecting the social field towards a new attractor basin.

The optimal regulatory intervention is the WKB tunnelling amplitude: the minimum J_{law} required to move the social field from the current equilibrium to the target equilibrium, accounting for the energy barrier height. Under-powered regulation (insufficient J_{law}) fails to move the system; over-powered regulation causes overshoot and unintended consequences.

This gives a formal criterion for regulatory adequacy: a regulation is adequate if and only if $|J_{\text{law}}| \geq T^{-1}$ where T is the WKB tunnelling amplitude between the current and target attractors.

The Rule of Law as Ergodicity

The rule of law — the principle that no person or entity is above the law — has a precise USF interpretation: it is the ergodicity condition on the social field.

A field is ergodic if every state accessible under the laws of physics is reachable from every other state given sufficient time. In legal terms: every social actor is subject to the same legal constraints; no actor has access to prohibited states that are closed to others.

Violations of the rule of law are violations of ergodicity: they create a two-tier attractor landscape where some actors have access to basins (corrupt practices, impunity) that are energetically accessible to them but legally prohibited to others. The topological signature of this violation is a broken symmetry in the constraint surface.

Legal Uncertainty and Attractor Fragmentation

When law is unclear or inconsistently applied, the attractor landscape fragments: multiple incompatible social equilibria coexist with no shared basin boundary. This is the USF model of legal uncertainty.

In a well-functioning legal system, the attractor landscape has a dominant cooperative equilibrium (law-abiding behaviour) connected by a clear energy gradient to all other states. In a system with high legal uncertainty, the landscape has multiple competing equilibria, none dominant, and transitions between them are unpredictable.

This gives a quantitative measure of legal certainty: the spectral gap of the constraint operator — the gap between the energy of the dominant legal equilibrium and the next local minimum. A large spectral gap means clear law; a small gap means ambiguous law.

Constitutional Law as Meta-Constraints

Constitutional law operates at a higher level than ordinary law: it constrains the space of permissible laws, not just the space of permissible behaviours. In USF terms, constitutional law is a meta-constraint on the operators $J_{\text{law}}(t)$ — a restriction on which regulatory interventions are permissible.

A constitutional provision is a constraint on the Hamiltonian operator H itself: it specifies which energy landscapes are reachable by ordinary legislation. A constitutional amendment changes the meta-constraint, expanding or restricting the space of permissible legal systems.

This explains the special difficulty of constitutional change: it requires modifying the meta-Hamiltonian, not just the current energy landscape. It is a second-order phase transition in the legal field.

Applications and Predictions

Regulatory design: The optimal regulation minimises the WKB tunnelling amplitude required to shift the social field to the target equilibrium, subject to constitutional meta-constraints.

Judicial review: A court reviewing regulatory action is checking whether $J_{\text{law}}(t)$ is consistent with the constitutional meta-constraint — whether the intervention respects the topological invariants.

Legal evolution: Common law develops by successive small deformations of the energy landscape (precedent), while statute changes it discontinuously (legislation). USF predicts that common law changes are path-dependent (hysteretic) while statutory changes can be discontinuous.

Prediction: Legal systems with higher spectral gap (clearer law) should show lower transaction costs, faster convergence to cooperative equilibria, and greater social trust — all measurable empirically.

Conclusion

Law is not arbitrary social convention. It is a field-theoretic technology for shaping the energy landscape of social attractor space. Rights are topological invariants. Regulation is volitional source-term engineering. The rule of law is ergodicity.

The method used to find this identification is documented in the Mathematical Co-identification paper. That method is now history. The structure stands.
